ZOHO CREATOR WORKFLOW ARCHITECTURE

Patient Appointment Booking - Automation Suite & Workflow Options

Aim

To build and document the automation layer for the **Patient Appointment Booking** system using Zoho Creator workflow options, covering form workflows, schedules, approvals, payments, blueprints, batch workflows, report workflows, and custom Deluge functions.

1. Objective

This experiment demonstrates the implementation of all eight Zoho Creator workflow options to automate the end-to-end patient care lifecycle, from initial appointment booking and check-in to consultation stage transitions, payment collection, and post-visit summary generation.

2. Workflow Mapping Overview

Workflow Type	Configuration / Purpose
Form Workflow	`On_Appointment_Submit_Workflow` - Triggers confirmation emails and SMS notifications to patients upon booking.
Schedule	`Daily_Appointment_Reminder_Schedule` - Sends daily automated morning reminders for scheduled consultations.
Approval	`Specialist_Consultation_Approval` - Manages doctor approval for specialized procedure/treatment requests.
Payment	`Consultation_Fee_Payment_Workflow` - Automates booking fee collection via online payment gateway integration.
Blueprint	`Patient_Care_Lifecycle_Blueprint` - Controls patient journey stages: Scheduled → Checked-In → In-Consultation → Completed.
Batch Workflow	`Bulk_Patient_Followup_Batch` - Runs nightly batch processing for post-consultation feedback & review triggers.
Report Workflow	`OnEdit_Recalculate_Billing_Workflow` - Recalculates total consultation and diagnostic fees upon report inline updates.
Function	`Generate_Prescription_Summary_Function` - Reusable Deluge script generating and emailing clinical visit summaries.

3. Detailed Workflow Configurations & Evidence

Form Workflow (On_Appointment_Submit_Workflow)

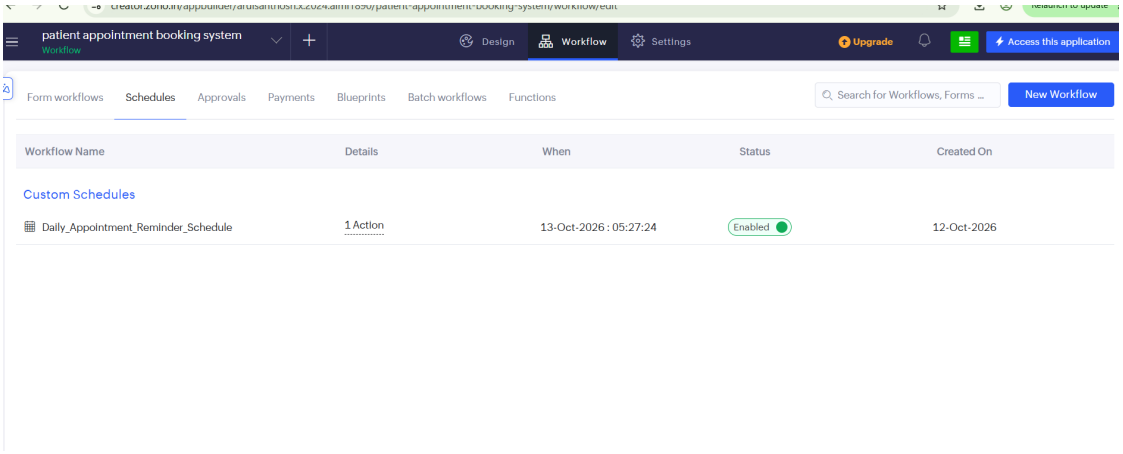
- **Purpose:** Automatically sends an email and SMS confirmation to the patient and alerts the duty reception cell upon new record creation in `Appointment_Booking_Form`.
- **Action:** Configured with a `Send Mail` action executed without conditions immediately after successful record creation.

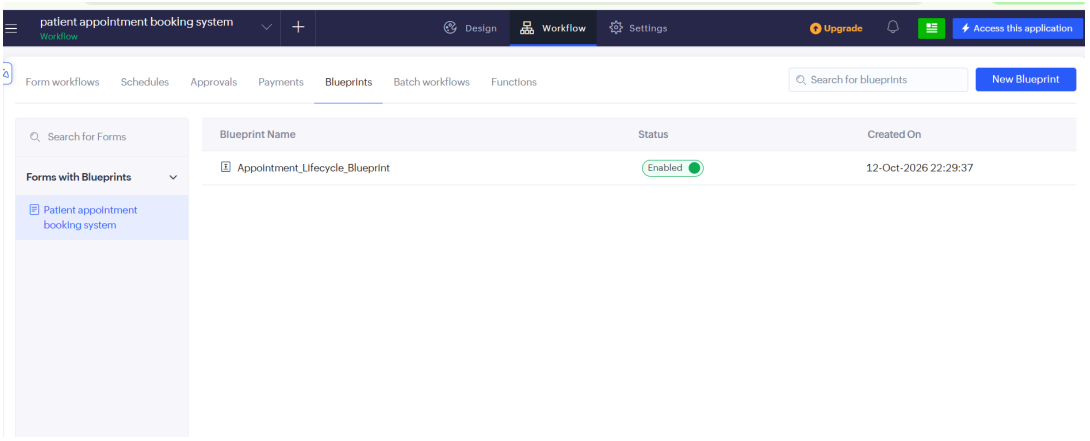
Schedule Workflow (Daily_Appointment_Reminder_Schedule)

- **Purpose:** Executes daily at 08:00 AM to fetch all appointments booked for the current date.
- **Action:** Automatically dispatches SMS alerts containing appointment time, doctor name, and clinic room number.

Approval Workflow (Specialist_Consultation_Approval)

- **Purpose:** Enforces specialist doctor approval for high-tier diagnostic procedures and surgeries.
- **Events:** Handles "On Start", "Level 1 Doctor Review", with conditional branching for Approve (schedules slot) and Reject (notifies patient with reason).



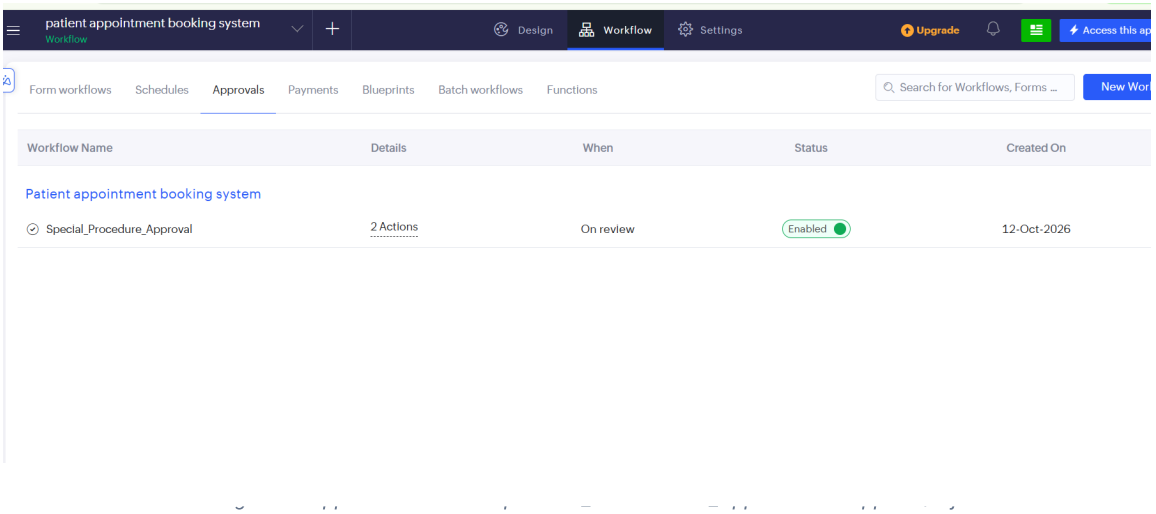


Blueprint (Patient_Care_Lifecycle_Blueprint)

- **Purpose:** Governs stage transitions for patient visits to maintain clinic operational discipline.
- **Stages & Transitions:** `Scheduled` → (Check-In) → `Checked-In` → (Start Consultation) → `In-Consultation` → (Complete Visit) → `Completed`.

Batch Workflow (Bulk_Patient_Followup_Batch)

- **Purpose:** Executes bulk updates for records in batches of 500 during off-peak hours (11:00 PM).
- **Deluge Integration:** Scans completed visits, evaluates follow-up dates, and generates recurring consultation tasks for chronic care patients.



Function Implementation (Generate_Prescription_Summary_Function)

- **Purpose:** Uses field link names (`Patient_Name`, `Doctor_ID`, `Diagnosis`, `Medication_List`) to fetch visit records, generate clinical documentation, and trigger automated PDF email delivery.

Result

The **Patient Appointment Booking Automation Suite** was successfully designed, configured, and documented using Zoho Creator's advanced workflow builder framework, seamlessly incorporating form triggers, schedules, approvals, payment workflows, blueprints, batch processing, report workflows, and custom Deluge functions.